

Professional Profile

NLP Researcher | Software Engineer | Network Infrastructure Engineer

	Nyanchoka Joel Otachi
Qualification	MSc. Computer Science (Candidate), Maseno University BSc. Computer Technology (Second Class Honours, Upper Division), Maseno University
Contact	joelotachi@gmail.com Phone: +254 113 368 690
Brief Bio	Nyanchoka Joel Otachi is a software engineer, NLP researcher, and network infrastructure specialist who builds, secures, and optimises the technical systems organisations depend on. He is currently pursuing an MSc in Computer Science at Maseno University and holds a BSc in Computer Technology (Second Class Honours, Upper Division) from the same institution. As a freelance full-stack developer, he delivers websites, e-commerce platforms, mobile apps, and custom billing systems tailored to each client's needs. In ICT and infrastructure roles with the ICT Authority (ICTA) and the Ministry of Agriculture and Livestock Development, he has maintained 99.5% network uptime and resolved 95% of helpdesk issues for 50+ staff within SLA timelines. As an NLP Research Associate at the Maseno Centre for Applied Artificial Intelligence (MCAAI), he works on synthetic data, dataset construction, and model fine-tuning for low-resource African languages. He is a native Ekegusii speaker and holds Cisco CCNA, JavaScript Essentials, Python Essentials, and IC3 certifications.

Core Skills	Software Engineering: Full-stack web development, UI/UX design (Figma), HTML, CSS, JavaScript, PHP, Laravel, Node.js, Next.js, Flutter, MySQL, PostgreSQL, Tailwind CSS, Bootstrap, API and M-Pesa integration, cloud hosting and deployment. Research and AI: Cross-lingual transfer learning, synthetic data and back-translation pipelines, dataset construction, model training, fine-tuning and evaluation for low-resource languages. Network Infrastructure: LAN/WAN design and administration, Cisco networking (CCNA), VLANs, VPN, VoIP, firewalls, CCTV and biometric systems, Microsoft 365 administration, backup and recovery.
Research Interests	Low-Resource NLP, Cross-Lingual Transfer Learning, Synthetic Data Generation, Dataset Construction, Intent Classification, and Model Fine-Tuning for African Languages.
Current Research	MSc Thesis: “An AfriBERTa-Based Cross-Lingual Transfer Learning Model for Intent Classification in Low-Resource Ekegusii.” Supervised by Dr. Lilian D.A. Wanzare and Dr. James Obohuma. MCAAI Projects: Building synthetic datasets and machine translation resources for low-resource Kenyan languages, and developing Small Language Models (SLMs) for indigenous Kenyan languages.
Attachment	Full CV: Available on request via joelotachi@gmail.com
Online Presence	Portfolio: joel-max-eng.github.io/Joel_Otachi GitHub: github.com/Joel-max-eng LinkedIn: linkedin.com/in/joel-otachi-79466738a